Live Demonstration Guide

CSCE 5300 | Project Group 7

Preparation

Open Terminal in the project root. Activate .venv. Verify Java and Spark are available. Close unrelated applications to reduce resource contention.

Suggested demonstration

Show the local clustering, partitioning, and global refinement modules. Use a small synthetic dataset for the demonstration. Explain the partition count, cluster count, convergence, and SSE.

Existing demonstration scripts

scripts/demo_partitioning.py
scripts/demo_parallel_kmeans.py
Confirm each script's command-line arguments and runtime before presenting.

Avoid

Do not rerun the full 10,000- or 50,000-record benchmark matrix during the live presentation.

Backup plan

Keep the existing experimental figures and verified CSV summaries available. If a live execution fails, explain the pipeline using those saved results.